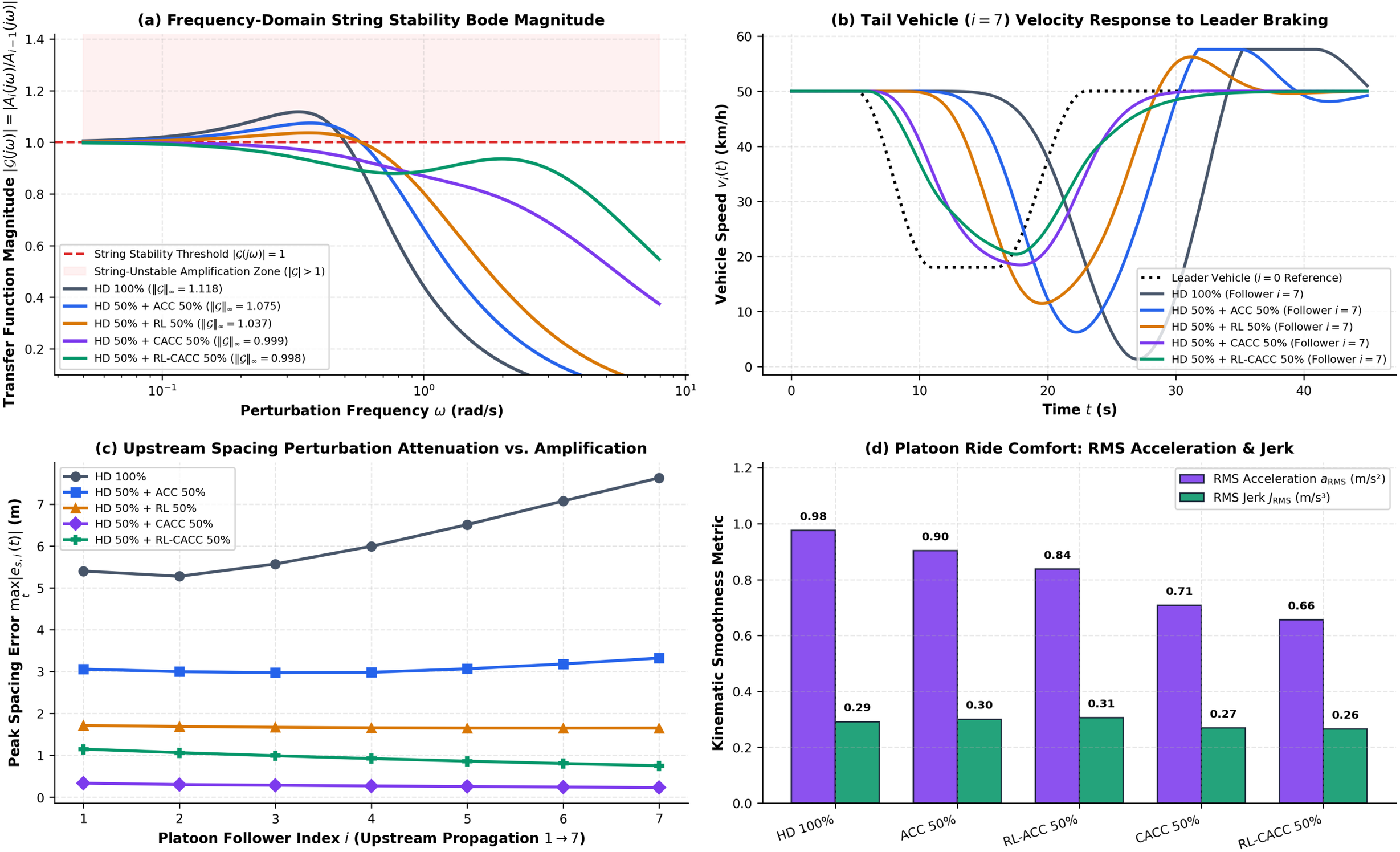

(a) Frequency-Domain String Stability Bode Magnitude
Transfer Function Magnitude $|\mathcal{G}(j\omega)| = |A_i(j\omega)/A_{i-1}(j\omega)|$
Perturbation Frequency ω (rad/s)
String Stability Threshold $|\mathcal{G}(j\omega)| = 1$
String-Unstable Amplification Zone ($|\mathcal{G}| > 1$)
HD 100% ($\|\mathcal{G}\|_\infty = 1.118$)
HD 50% + ACC 50% ($\|\mathcal{G}\|_\infty = 1.075$)
HD 50% + RL 50% ($\|\mathcal{G}\|_\infty = 1.037$)
HD 50% + CACC 50% ($\|\mathcal{G}\|_\infty = 0.999$)
HD 50% + RL-CACC 50% ($\|\mathcal{G}\|_\infty = 0.998$)
(b) Tail Vehicle ($i = 7$) Velocity Response to Leader Braking
Vehicle Speed $v_i(t)$ (km/h)
Time t (s)
Leader Vehicle ($i = 0$ Reference)
HD 100% (Follower $i = 7$)
HD 50% + ACC 50% (Follower $i = 7$)
HD 50% + RL 50% (Follower $i = 7$)
HD 50% + CACC 50% (Follower $i = 7$)
HD 50% + RL-CACC 50% (Follower $i = 7$)
(c) Upstream Spacing Perturbation Attenuation vs. Amplification
Peak Spacing Error $\max_t |e_{s,i}(t)|$ (m)
Platoon Follower Index i (Upstream Propagation $1 \to 7$)
HD 100%
HD 50% + ACC 50%
HD 50% + RL 50%
HD 50% + CACC 50%
HD 50% + RL-CACC 50%
(d) Platoon Ride Comfort: RMS Acceleration & Jerk
Kinematic Smoothness Metric
RMS Acceleration a_{RMS} (m/s²)
RMS Jerk J_{RMS} (m/s³)
0.98
0.29
0.90
0.30
0.84
0.31
0.71
0.27
0.66
0.26
HD 100%
ACC 50%
RL-ACC 50%
CACC 50%
RL-CACC 50%